

	Work Instruction	Doc. Revision	01
	AXI V810 X-Axis Min Travel Stroke Optimization Procedure	Effective Date	16 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 Jul 2015	First Release	CHIN CHEE YAN
01	16 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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X Axis Travel Stroke Checking

- Go to Digital I/O tab and initialize as show in Figure 1.

Services → Subsystem Status & Control → Digital I/O → Initialize

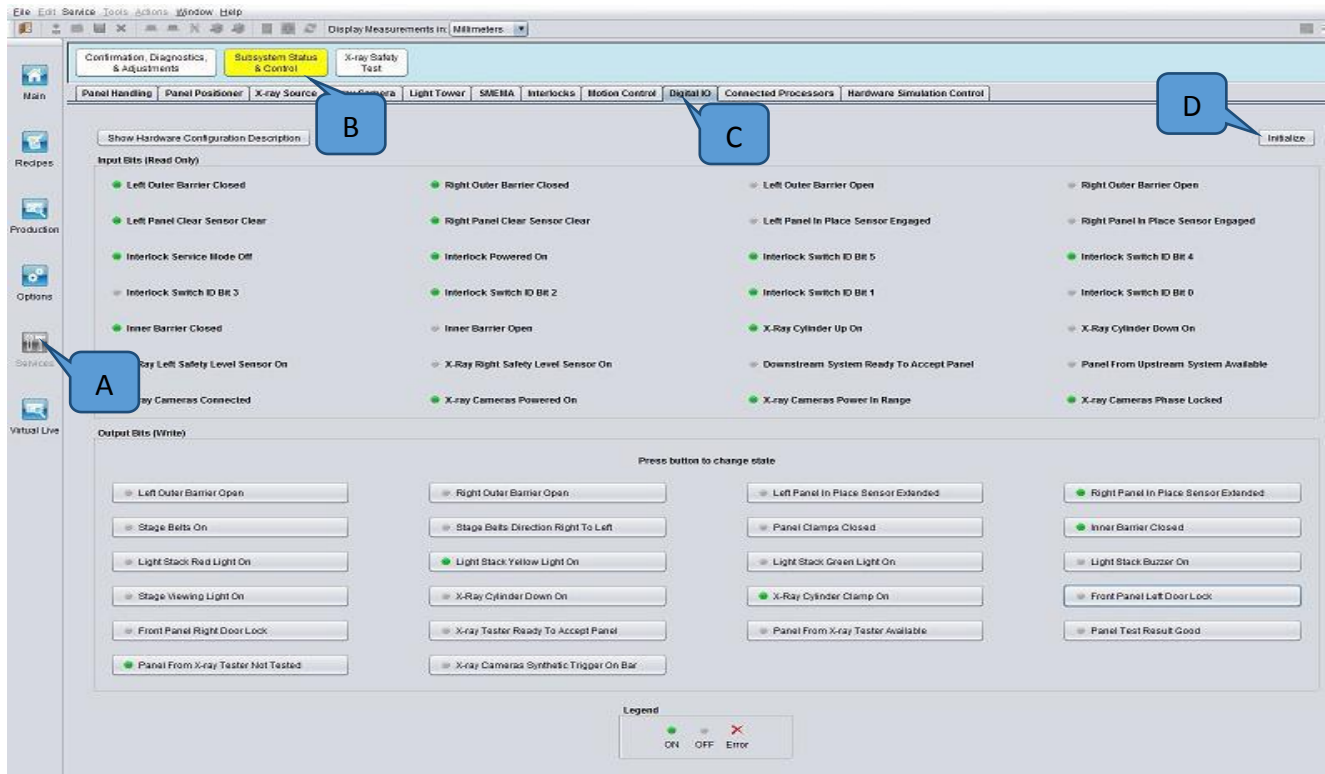


Figure 1 Digital IO

- Go to Panel Handling tab and initialize as show in Figure 2. Change display measurement unit to MM

Services → Subsystem Status & Control → Panel Handling → display unit to MM → Initialize

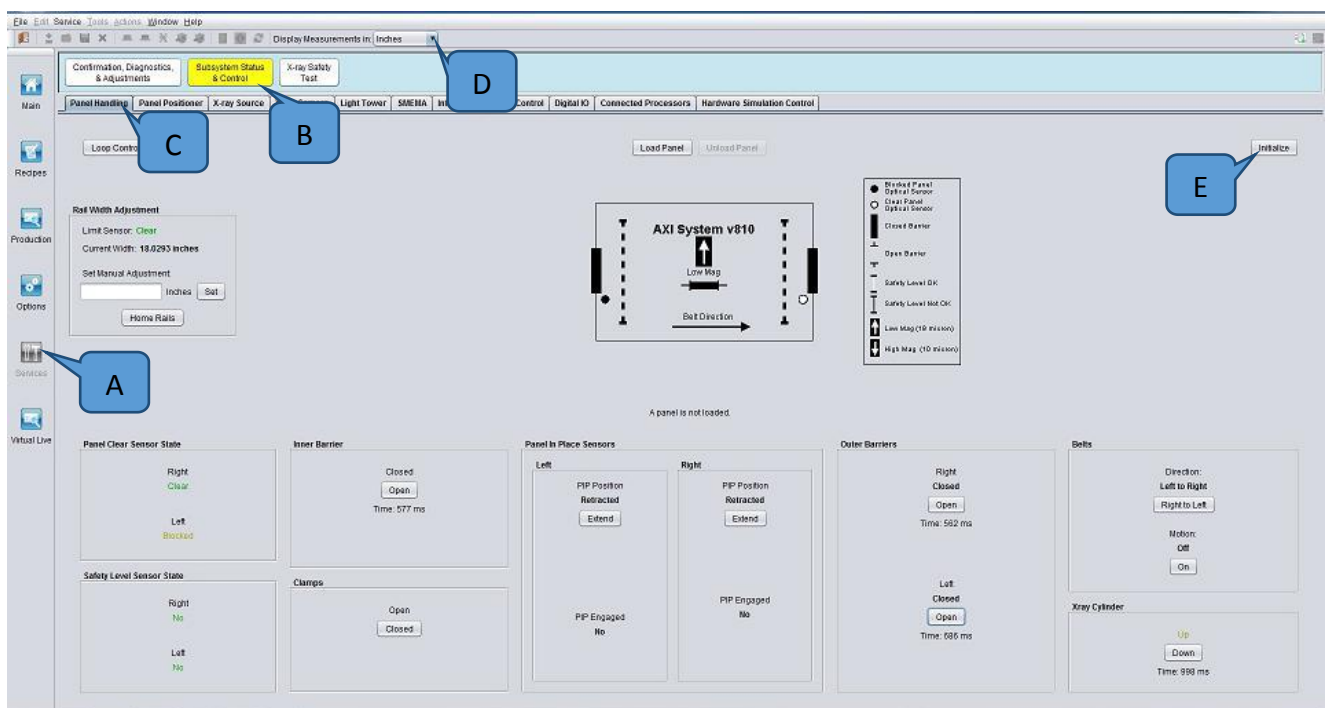


Figure 2 Panel handling

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- Go to Panel Positional tab, click initialize and disable the Motion X&Y Axis. Turn off X ray and Open the front access door.

Services → Subsystem Status & Control → Panel Positional → Initialize → Disable X and Y Motion → Turn off X-ray Source Operator Console → Open Front Access Door

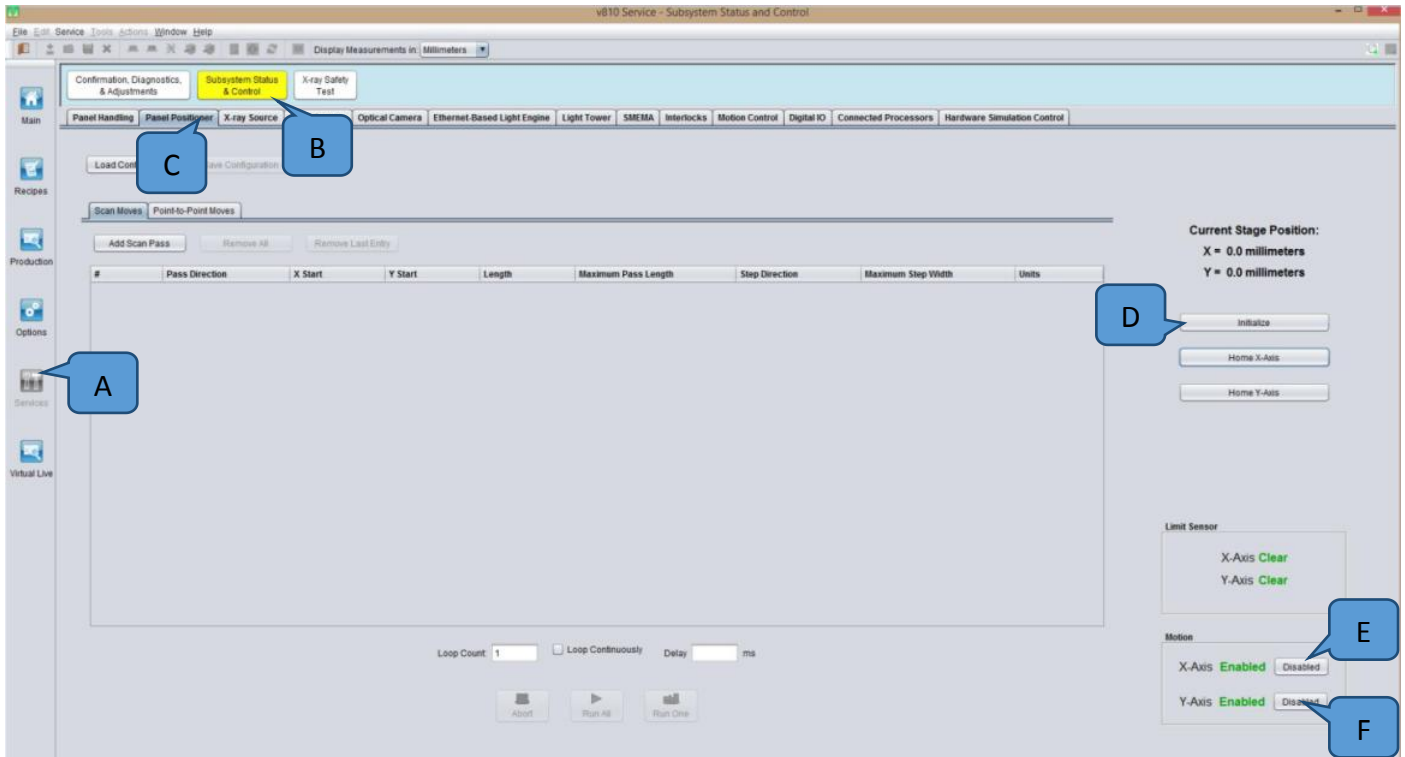


Figure 3 Panel Positioner

- Move the stage toward to Left Outer Barrier until the stage is collide with X axis Minimum hard stop.

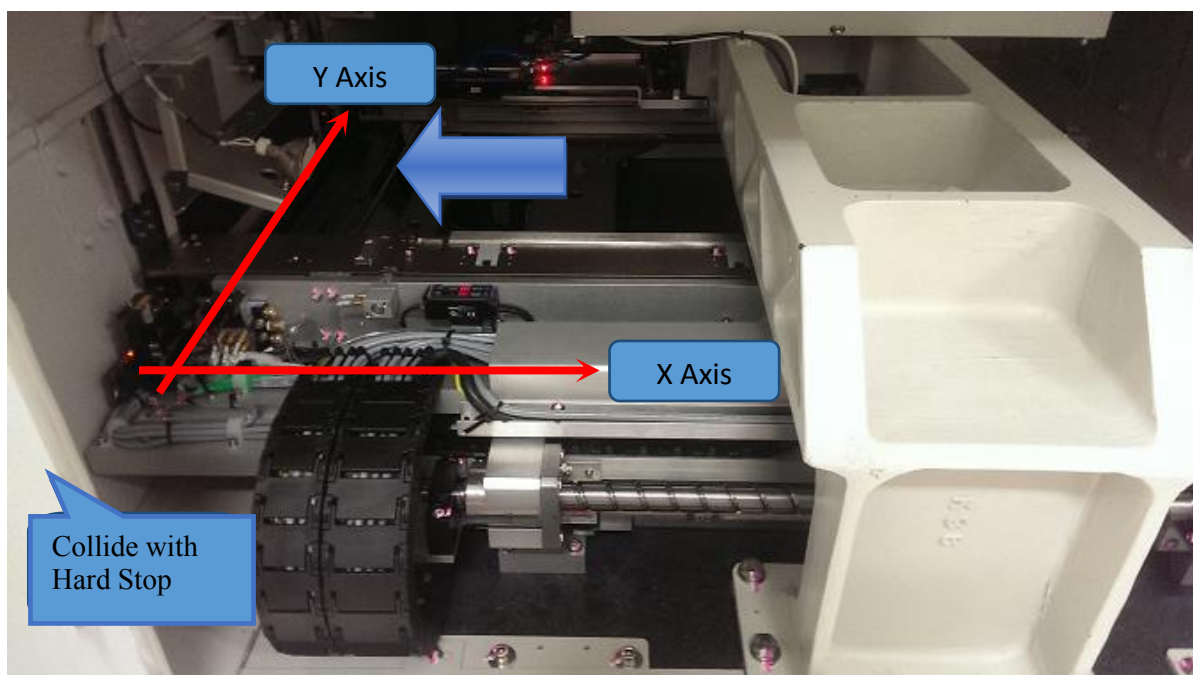


Figure 4 X Axis Minimum hard stop

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5. Check on the update value at Current Stage Position (Panel Handling tab).

- Spec must within **-3.00 mm ~ -3.30 mm**.

- During **stage Home position**, Limit Sensor show **Clear**, when move to **- 1 mm**, will start show **Block** until the end collide with left hard stop.

Example#*This will depend to the actual value show at existing machine.*

X= -6.181mm and need move both sensor toward to machine left side approximately 3 mm to optimize min travel stroke.

Red sign assume as starting point.

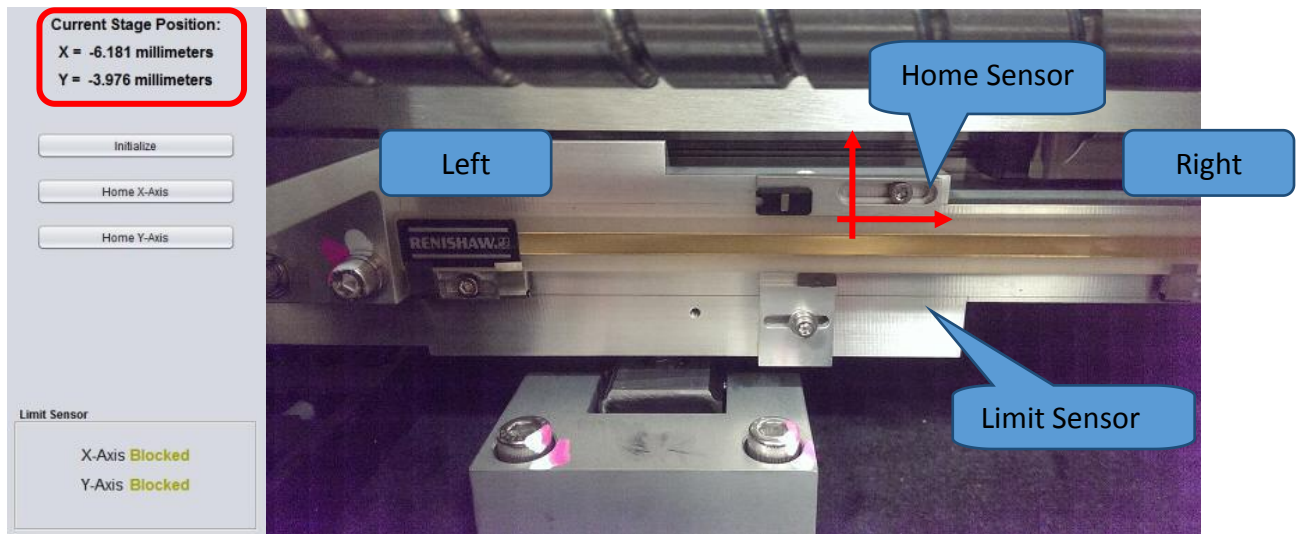


Figure 5 Home Sensor & Limit Sensor

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X Axis Optimization

Home sensor adjustment

- Perform a marking at Home Sensor bracket.
- Loosen the M3 screw as shown below in the Figure 6.

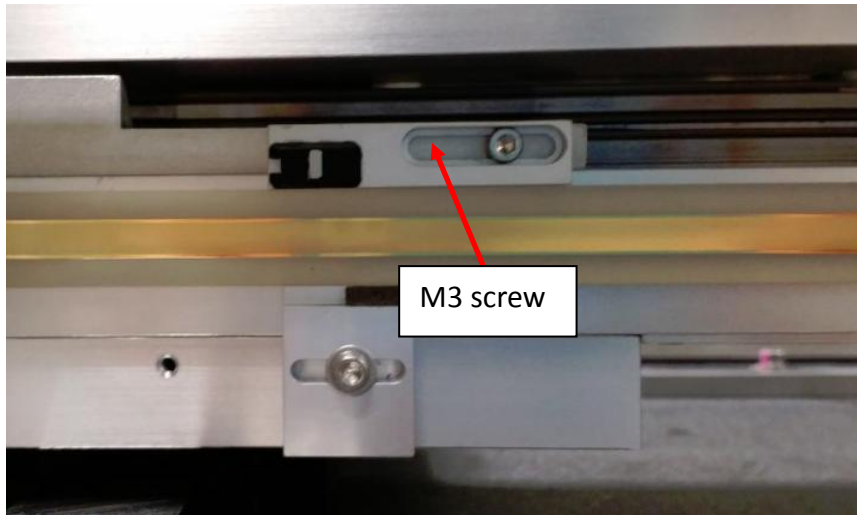


Figure 6 Home Sensor M3 Screw

- Adjust the X-axis as below table.

Current stage position after initialize the Panel Positioner	Direction of the home sensor need to be adjusted
Less than -3.00mm	Towards the left outer barrier side
More than -3.30mm	Towards the right outer barrier side

- Once home sensor adjustment complete, tighten M3 screw as shown in Figure 6.
- Push the X axis & Y axis stage to the middle of the machine and close front access door.
- Initialize at Panel Positioner Tab.
- Open up front access door.
- Move the X axis stage toward to Left Outer Barrier until the stage is collide with X axis Minimum hard stop.
- Check on the update value at Current Stage Position (Panel Handling tab).

*Spec must within **-3.00 mm ~ -3.30 mm.***

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15. Repeat Step 7-14 until X axis home sensor is within the range from -3.00mm to -3.30mm.
16. In the Panel Positioner tab, change the **Display Measurement** to **Nanometres** and click Initialize.
After initialization complete, disable the motion for the X-axis and Y-axis.
17. Open the front access door and push the X axis towards right outer barrier direction until it reach to X Axis Maximum Hard Stop.



Figure 7 X Axis Maximum Hard Stop

18. Verify X axis Actual Max travel stroke (Display at Panel Positioner `s Current Position) where MUST more than Command Distance where refer to below Machine Model.

Machine Model	Command Distance (NM)	X axis Actual Max Travel Stroke (NM)
V810 S2	501650000	Required to be more than Command Distance Value
V810 XXL S2	859941200	Required to be more than Command Distance Value
V810 S2EX	532650000	Required to be more than Command Distance Value

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Limit sensor adjustment

19. Close the V810 software.
20. Login to the Atom PC and enter the computer name as shown below in Figure 8 (S1 Series) and Figure 9. (S2 Series)

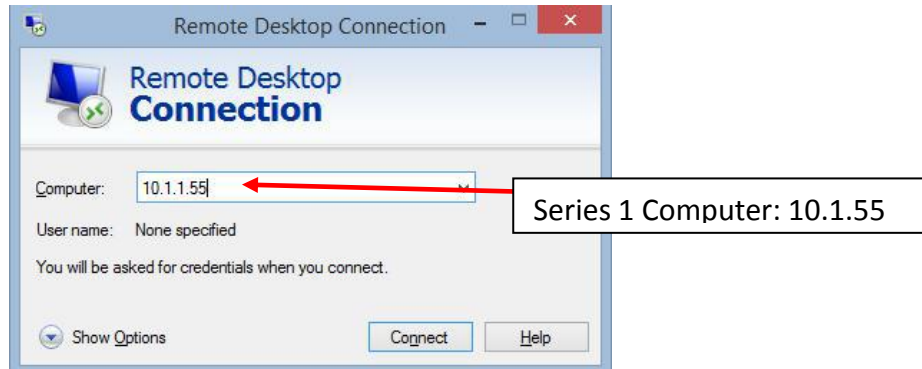


Figure 8 Series 1

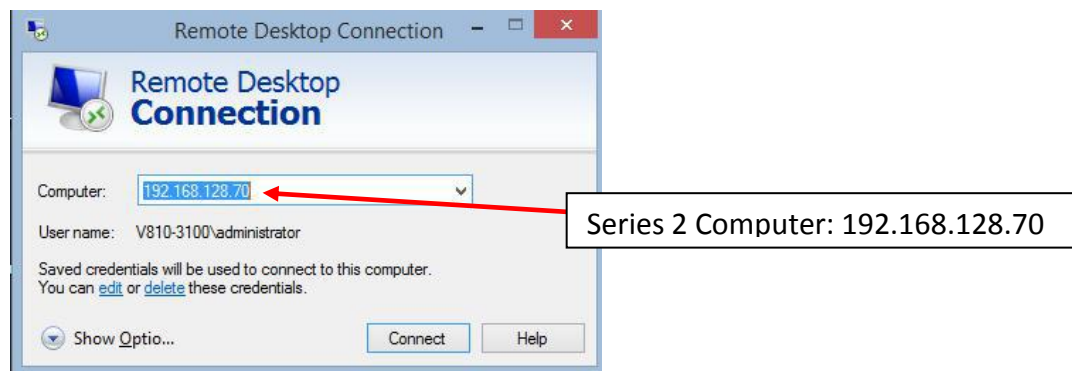


Figure 9 Series 2

21. Enter the password as "Please!" as shown in Figure 10.

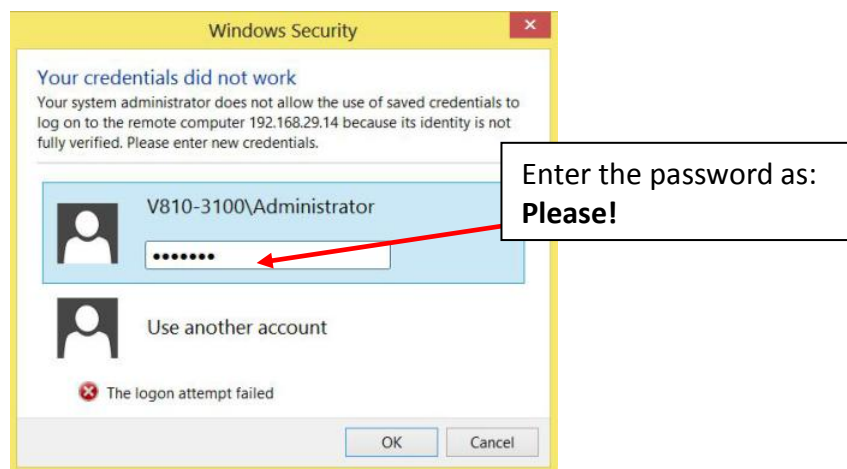


Figure 10 Password for Atom PC

22. Go to the following directory: C:\Program Files\AXI System\v810\REV\bin and open the MC_XMP_NT.EXE as shown in Figure 11.

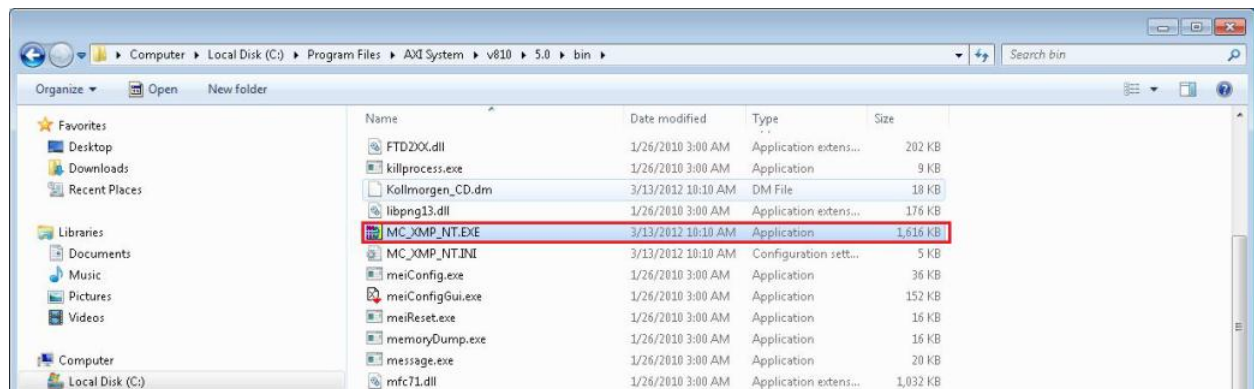


Figure 11 MC_XMP_NT.EXE

23. Ensure this 3 tab is open in the Motion Console tab as shown in Figure 12.

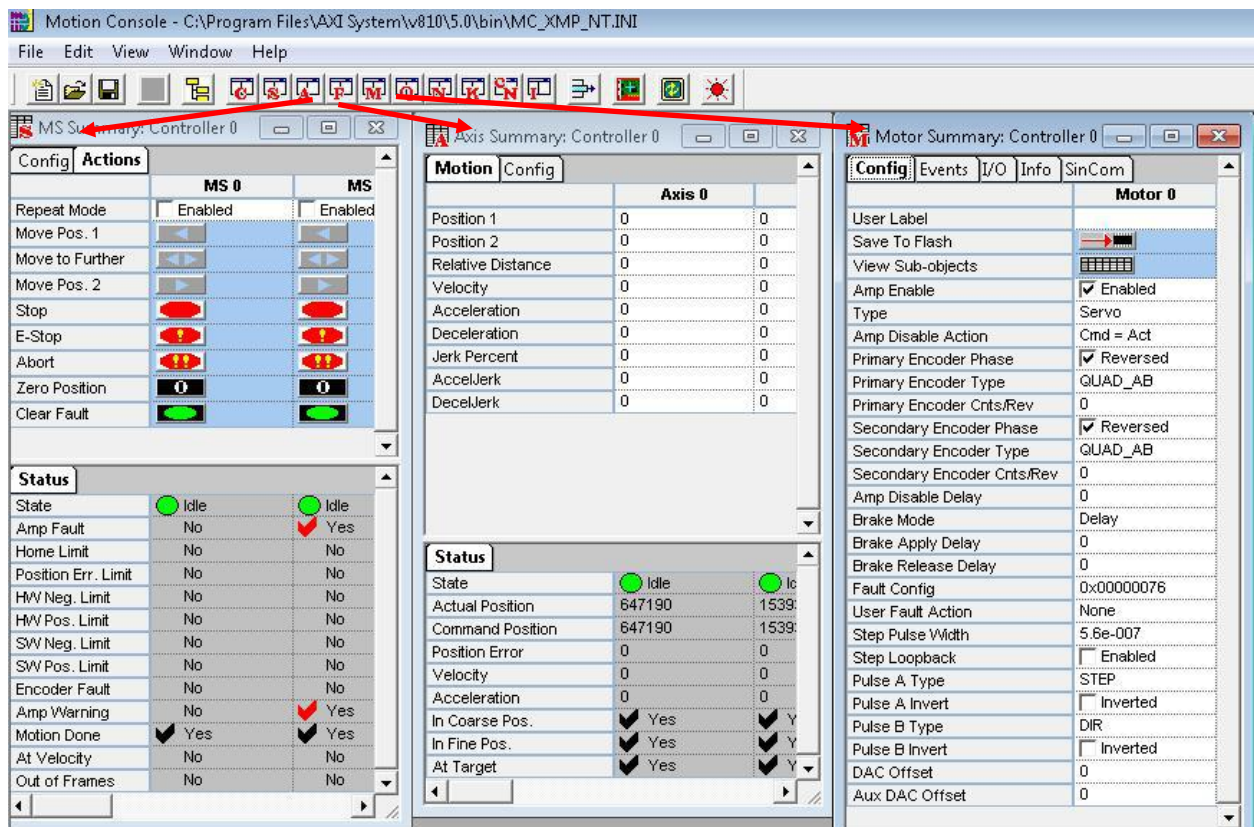


Figure 12 Motion Control Tab

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24. Close the X-ray source and open the front access door.

25. Push the stage towards the minimum hard stop (left outer barrier) as shown in Figure 13.

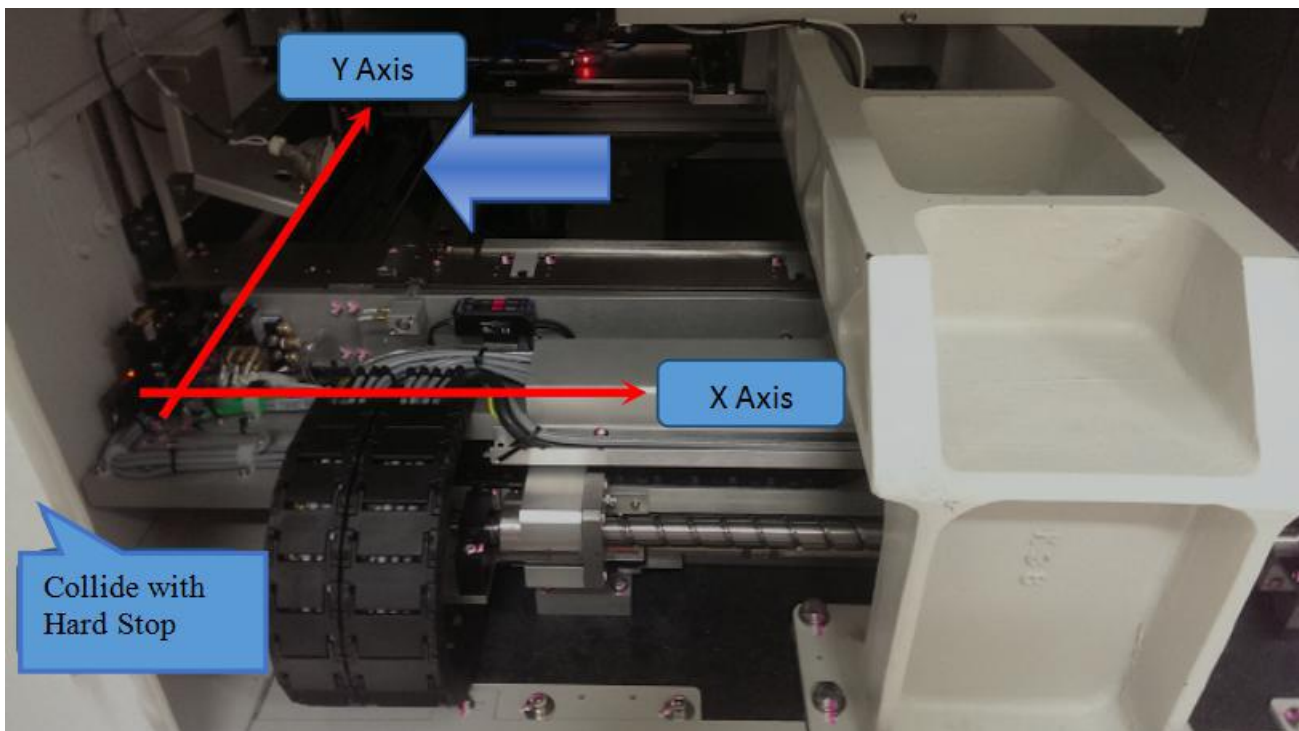


Figure 13 Minimum hard stop

26. Reset all MS0, MS1 and MS2 at the MS Summary (Motion Supervisor Summary) as shown in Figure 14. Once performed the reset for all the axis, the actual position for MS0, MS1 and MS2 will become zero as shown in Figure 15.

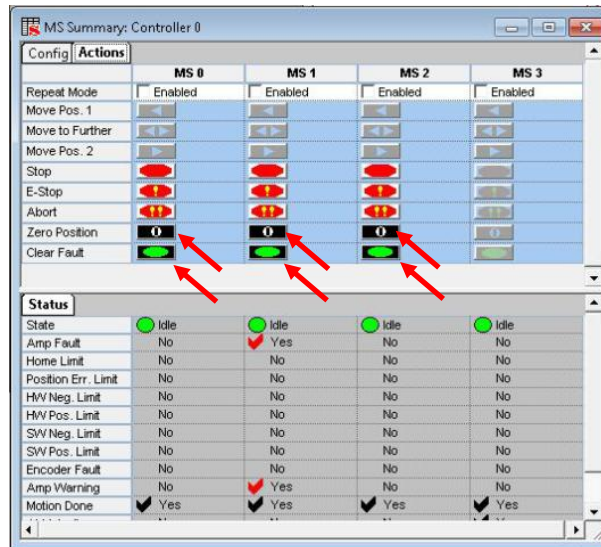


Figure 14 MS Summary (Motion Supervisor Summary)

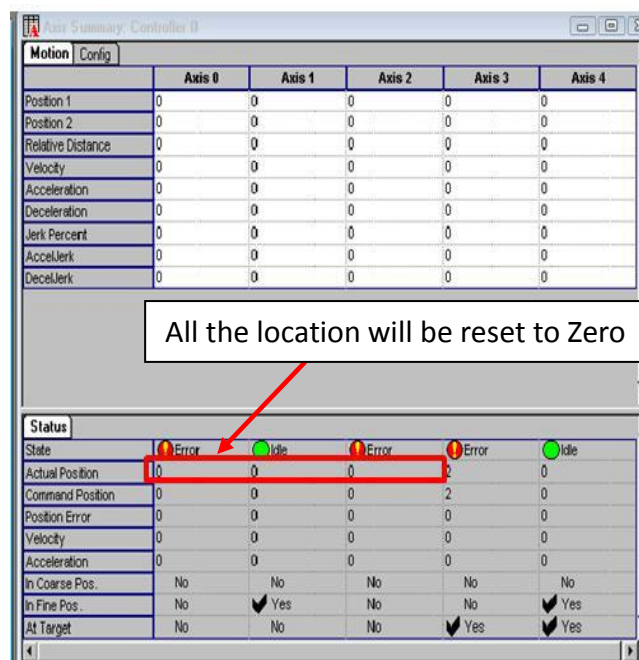


Figure 15 Axis Summary (Actual position)

27. Gently rotate the coupling for the X-Axis stage as shown in the Figure 16 until the Actual Position and Command Position in Figure 17 become 5990.

Remarks: Acceptable range for the Actual Position and Command Position will be from 5970~5990.

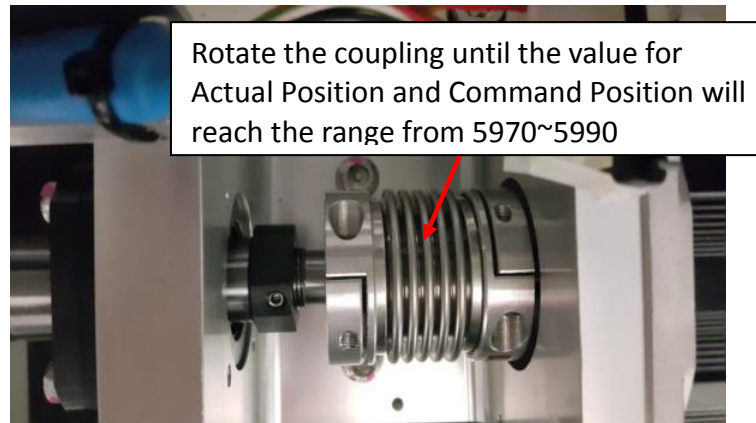


Figure 16 X-axis coupling screw

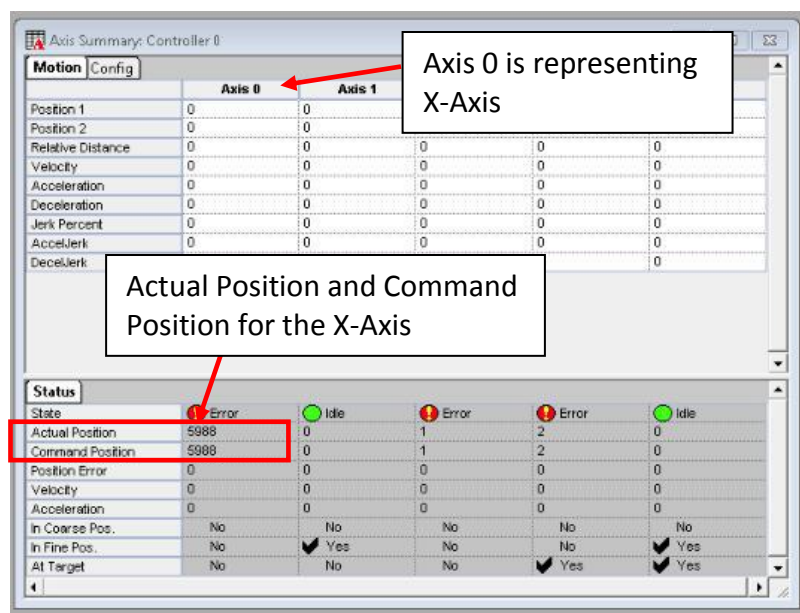


Figure 17 Actual Position and Command Position

28. Loosen the M3 screw of the limit switch as shown in Figure 18.

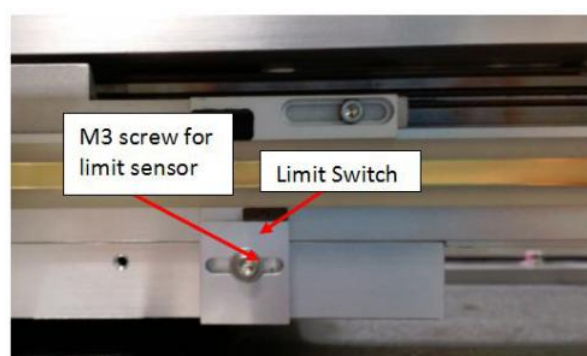


Figure 18 Limit Sensor for X-Axis

29. Adjust the limit switch shown in Figure 18 until the Home Sensor status become “High” at the Motor Summary as shown in Figure 19 and Figure 20.



Figure 19 Motor Summary (Home Sensor)

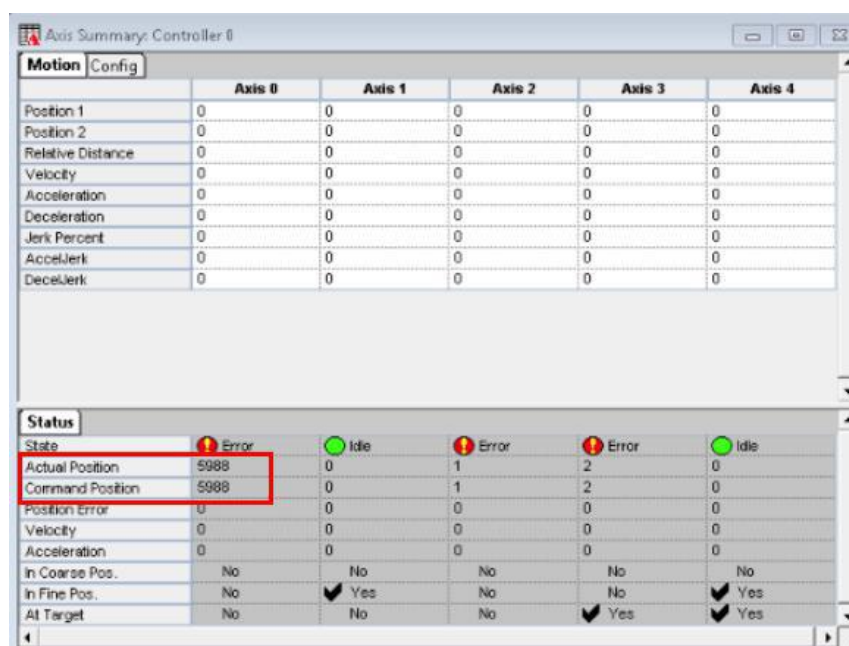


Figure 20 Axis Summary (Home Sensor)

30. Once the limit switch condition is acquired, tighten back the M3 screw for the limit sensor screw as shown in Figure 18.
31. Pull the X-Axis stage to the minimum hard stop.
32. Reset again the Axis 0 become zero as shown in Figure 21 and ensure the actual position for the Axis 0, Axis 1 and Axis 2 become zero as shown in Figure 22.

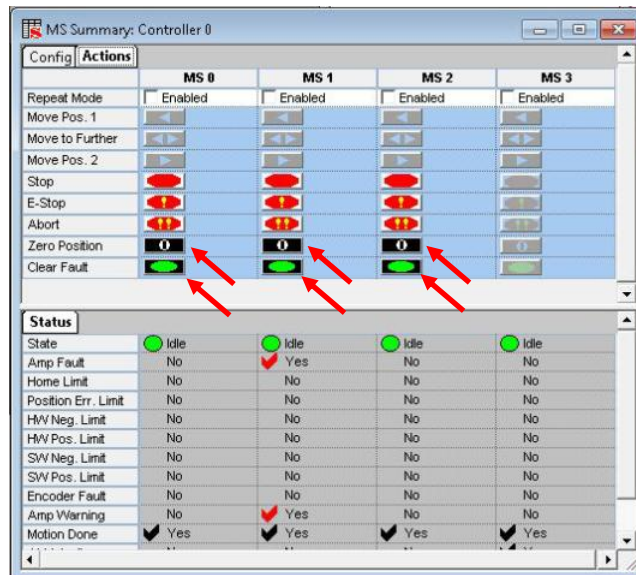


Figure 21 MS Summary

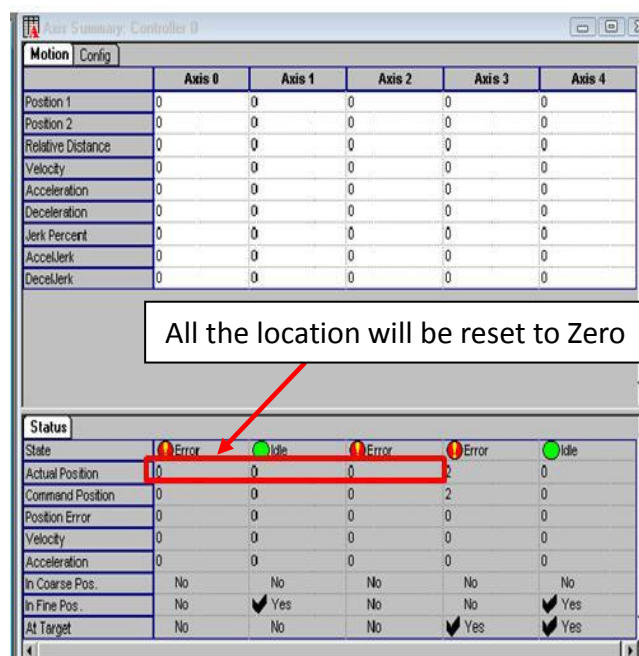


Figure 22 Axis Summary (Actual position)

33. Slowly rotate the X-axis coupling until see the status become “Low” at indicated at the “Home Sensor” as shown in Figure 23.

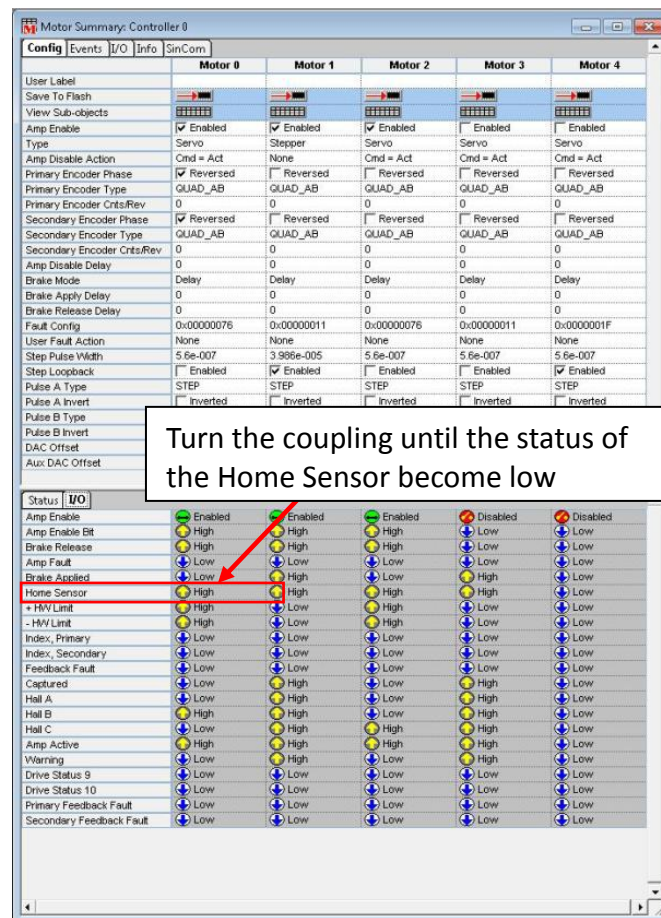


Figure 23 Motor Summary

34. Once the status becomes “Low”, X-axis stage should not move anymore.

35. Refer to the actual (From Axis Summary) and verify the limit sensor position. May refer to below table for counter reference.

X axis Actual position (Axis 0 at Axis summary)	Expected result at Motor 0 (Motor summary)
5000~5200	Home Sensor Low
5700~5990	Home Sensor High

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36. Transition signal from low to high will happen at the range of 5000-6000 counts. If the result is not able to achieved as in Table 1 (Low at 5000-5200 and High at 5700-5990), please do repeat from Step 23-35.

37. Once complete, exit the Motion Console software.

38. Push the XY-stage to the middle of the machine and close the front access door.

39. Open the V810 software. Go to the Services->Subsystem Status & Control-> Digital IO tab and Click Initialize as shown in Figure 24.

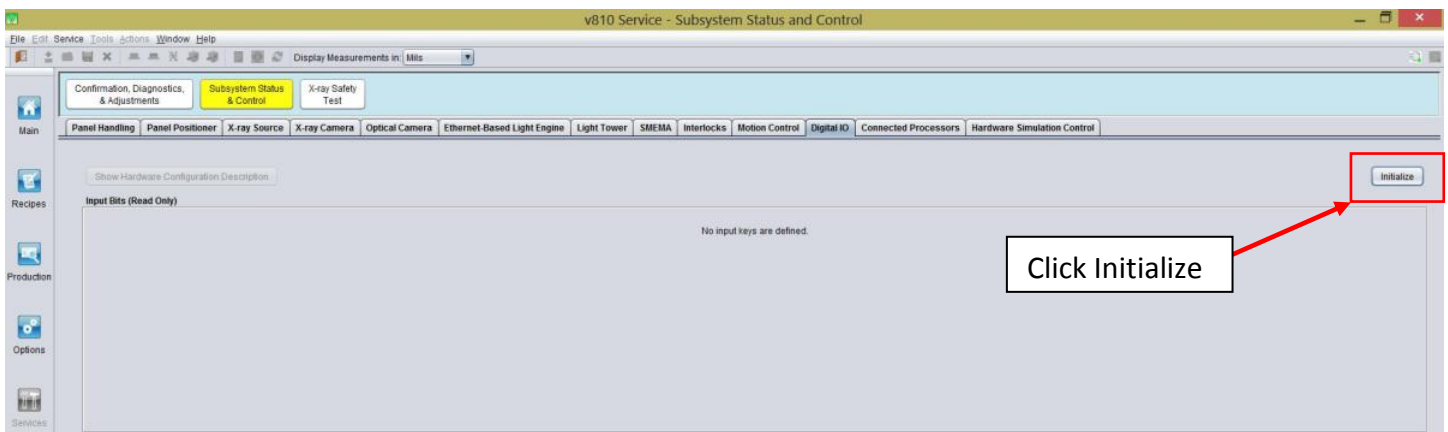


Figure 24 Digital IO

40. Go to the Panel Positioner tab and click Initialize. The limit sensor status for the X-axis should be clear. Disable the Motion for the X-axis and Y-axis as shown in Figure 25. Close the X-ray source and open the front door.

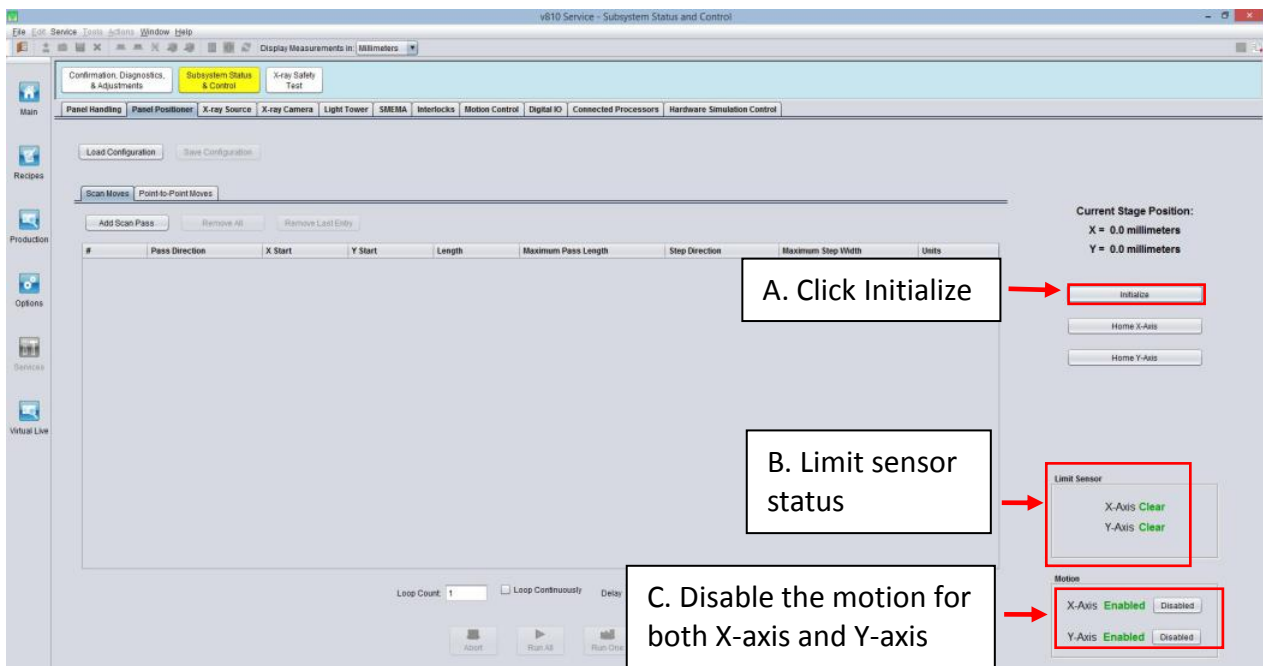


Figure 25 Panel Positioner

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41. Push the stage to the minimum hard stop (left outer barrier side) and status of the X-axis limit sensor should be indicated as “Blocked” as shown in Figure 26.

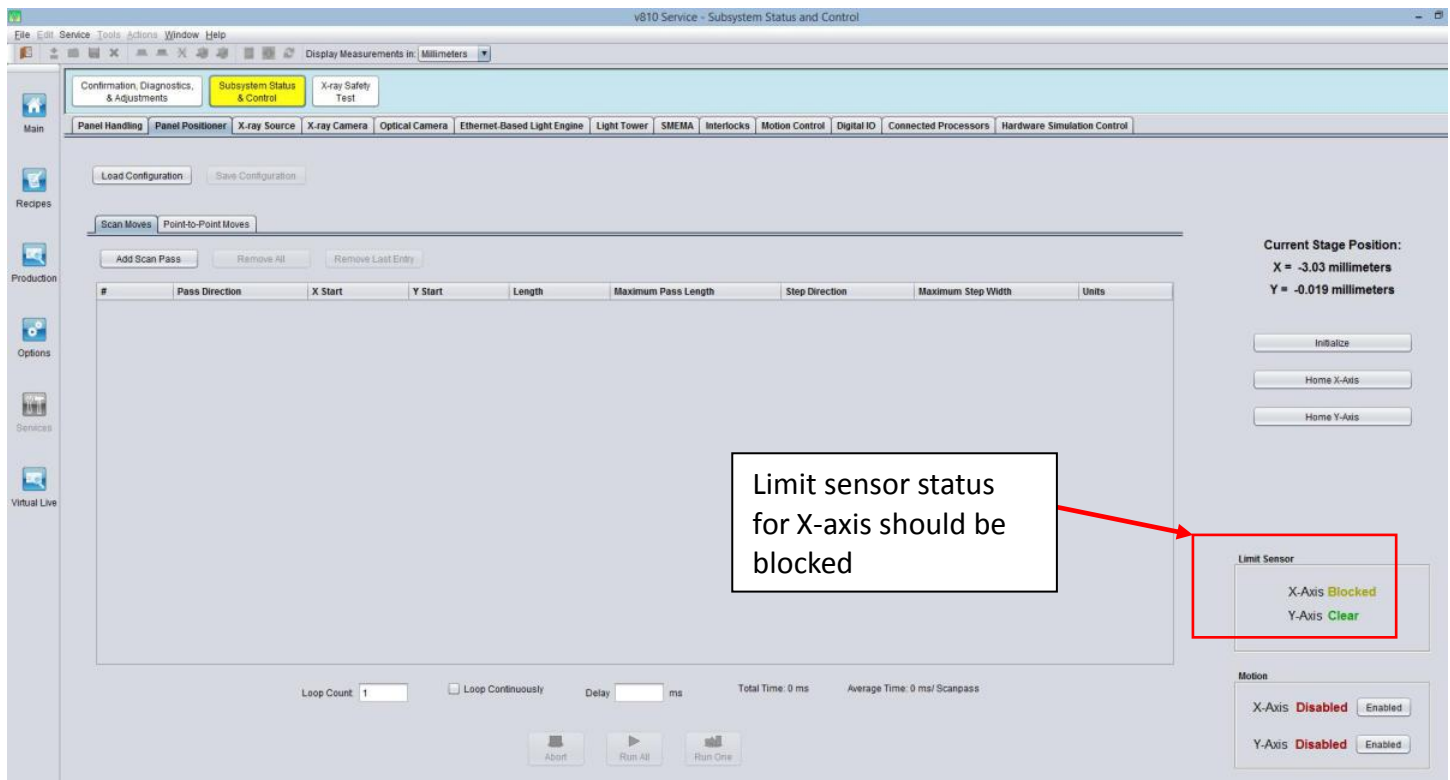


Figure 26 X-Axis Limit Sensor

42. If verification not be able to achieved, repeat Step 23-41.